TEXORA · LECTURE SHEET

Thermal Power Plants

A textile mill runs on two energy streams — electricity for drives and steam for processing. Both come from thermal plant. This sheet covers the basic principles, the cycles used, and the four plant families: steam, diesel, gas and combined cycle, with regeneration.

TE-4105 · INDUSTRIAL UTILITIES · TOPIC 2 · LEVEL 4 TERM 1

1. Basic principles

A thermal power plant converts the chemical energy of a fuel into electrical energy through heat and mechanical work:

Fuel chemical energy → heat → mechanical work (turbine/engine shaft) → electricity (generator)

Every such plant is a **heat engine** working between a hot source and a cold sink, so its efficiency is limited by thermodynamics: heat must be rejected to the condenser or exhaust, and no plant converts all fuel heat into work. The measure used throughout this sheet is:

Thermal efficiency η = net work output ÷ heat supplied

Common fuels are coal, natural gas, furnace oil and diesel. In Bangladesh, grid and captive mill plants are dominated by natural gas, with coal, oil and diesel also in use; large textile groups commonly run gas-engine or gas-turbine captive plants (see Topic 4 on cogeneration).

2. Cycles used

CYCLE	WORKING FLUID / MACHINE	USED IN
Rankine cycle	Water/steam; boiler + turbine + condenser + pump	Steam power plant
Diesel (and Otto) cycle	Air/fuel burned inside a reciprocating engine cylinder	Diesel power plant; gas engines
Brayton (Joule) cycle	Air/gas; compressor + combustor + gas turbine	Gas power plant
Combined cycle	Brayton topping + Rankine bottoming (via heat-recovery steam generator)	Modern gas-based stations

3. The Rankine cycle and the steam power plant

The ideal Rankine cycle has four processes:

1. **1 → 2 Pumping:** liquid water in the condenser is pumped to boiler pressure (small work input).
2. **2 → 3 Heat addition:** water is heated, evaporated and often superheated at (nearly) constant pressure in the boiler.
3. **3 → 4 Expansion:** steam expands in the turbine, producing shaft work; pressure and temperature fall.
4. **4 → 1 Heat rejection:** exhaust steam condenses at constant pressure in the condenser, rejecting latent heat to cooling water.

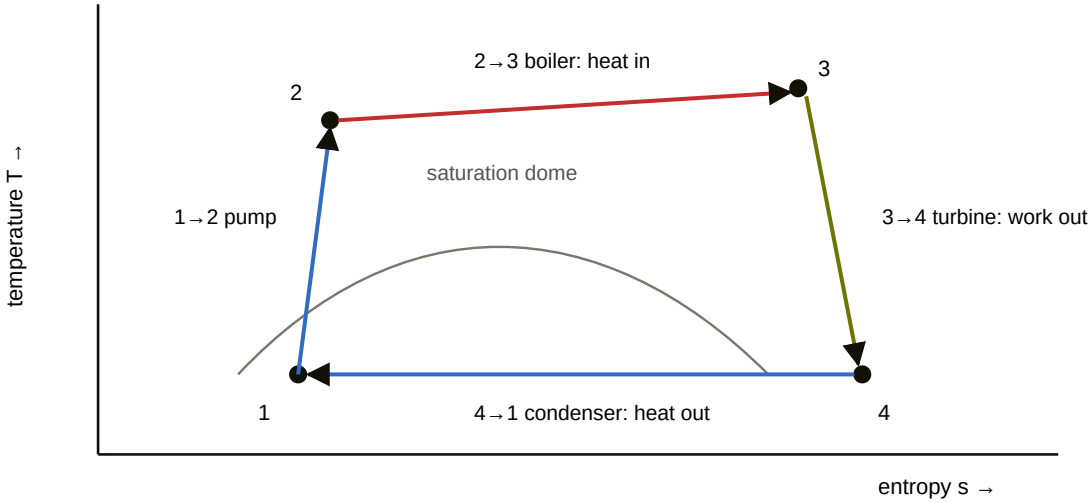


Figure 1 — T–s diagram of the Rankine cycle. 1 → 2 pumping, 2 → 3 constant-pressure heat addition in the boiler, 3 → 4 expansion in the turbine, 4 → 1 constant-pressure condensation. Point 3 includes superheat in a real plant.

WORKED EXAMPLE — RANKINE-CYCLE EFFICIENCY

Turbine inlet enthalpy h_3 = 3,200 kJ/kg; turbine exhaust h_4 = 2,200 kJ/kg; feedwater entering the boiler h_2 = 190 kJ/kg. Neglect pump work (so $h_1 \approx h_2$ in the denominator as given).

Turbine work = $h_3 - h_4$ = 3200 − 2200 = 1,000 kJ/kg.

Heat supplied = $h_3 - h_2$ = 3200 − 190 = 3,010 kJ/kg.

η = 1000 / 3010 = 0.332 ≈ **33.2%**.

Reading: of every 100 units of heat added in the boiler, about 33 become turbine work and about 67 are rejected in the condenser — which is exactly the heat a cogeneration plant tries to use instead (Topic 4).

Steam plant layout: fuel store and handling → boiler (Topic 3) → superheater → steam turbine → generator; exhaust → condenser → condensate pump → feed heaters → boiler feed pump → boiler. Auxiliaries include cooling tower (Topic 1), water treatment, ash handling (coal) and flue-gas cleaning. Steam plants suit large, steady outputs and can burn low-grade fuels, but start slowly and need large water supplies.

4. Regeneration

Regeneration (regenerative feed heating) improves the Rankine cycle by bleeding a fraction of steam part-way through the turbine to heat the feedwater before it enters the boiler. The bled steam gives up heat that would otherwise be rejected in the condenser, and the boiler receives warmer feedwater, so less fuel heat is needed per unit of work — efficiency rises. The bled steam does slightly less turbine work, but the fuel saving is larger, so the net effect is positive. Large steam plants use several feed heaters in series; textile boiler houses apply the same idea in simple form with a hot-well, deaerator and economiser (Topic 3).

5. Diesel power plant

A diesel plant burns diesel oil inside reciprocating engine cylinders (Diesel cycle: adiabatic compression, constant-pressure heat addition, adiabatic expansion, constant-volume heat rejection). The engine drives a generator directly.

- **Main parts:** engine, fuel injection system, air intake and supercharger, cooling-water system, lubricating-oil system, exhaust silencer, alternator and switchgear.
- **Advantages:** quick start, high part-load efficiency, compact, needs little water; ideal standby and peak duty.
- **Disadvantages:** fuel is expensive per kWh, capacity per unit is limited, maintenance is frequent and skilled.

Mills keep diesel sets as emergency standby for lighting, boilers' auxiliaries and safe machine stoppage during grid failure.

6. Gas power plant

Two forms are used. (a) **Gas-engine plant:** a reciprocating engine burning natural gas (Otto-type cycle), common as captive plant in Bangladeshi textile groups because it starts fast and follows load. (b) **Gas-turbine plant (open Brayton cycle):** compressor → combustion chamber → turbine, with compressor and generator on the same shaft line.

- **Advantages (turbine):** very compact for its output, fast start, low cooling-water need, clean exhaust with gas fuel.
- **Disadvantages (simple cycle):** efficiency falls sharply at part load and in hot weather; a large share of turbine work drives the compressor.

7. Combined cycle

A **combined-cycle** plant stacks the two cycles: the gas turbine (Brayton) is the topping cycle, and its hot exhaust — still containing much usable heat — passes through a **heat-recovery steam generator (HRSG)** that raises steam for a steam turbine (Rankine) as the bottoming cycle. One fuel supply thus produces power twice. Combined-cycle plants achieve the highest thermal efficiencies of the conventional thermal plants (modern large units are commonly quoted above 50%, illustrative textbook range), use less cooling water per kWh than a steam-only plant, and are the standard choice for large new gas-based stations.

FEATURE	STEAM PLANT	DIESEL PLANT	GAS TURBINE (SIMPLE)	COMBINED CYCLE
Cycle	Rankine	Diesel	Brayton	Brayton + Rankine
Typical unit size	Large	Small–medium	Medium–large	Large
Start-up	Slow (hours)	Fast (minutes)	Fast	Medium
Best duty	Base load	Standby / peak	Peak / mid	Base load
Fuel flexibility	High (coal, oil, gas, biomass)	Low (diesel oil)	Gas / light oil	Mainly gas
Cooling-water need	High	Low	Very low	Moderate
Relative efficiency	Moderate	Good at small scale	Low–moderate alone	Highest of the four

Glossary

Rankine cycle — the vapour power cycle of pump, boiler, turbine and condenser. **Brayton cycle** — the gas-turbine cycle of compression, combustion and expansion. **Regeneration** — heating feedwater with steam bled from the turbine to raise cycle efficiency. **HRSG** — heat-recovery steam generator; a boiler heated by gas-turbine exhaust. **Topping / bottoming cycle** — the higher-temperature and lower-temperature cycles of a combined plant. **Condenser** — heat exchanger condensing exhaust steam with cooling water. **Captive plant** — a plant owned by an industry for its own supply.

Key points

• All thermal plants are heat engines: η = net work ÷ heat supplied; heat rejection is unavoidable.
• Rankine example: (3200 − 2200)/(3200 − 190) = 1000/3010 ≈ 33.2%.
• Regeneration bleeds turbine steam to heat feedwater and raises Rankine efficiency.
• Diesel = fast, small, costly fuel — standby duty. Gas turbine = compact, fast, weaker alone. Combined cycle = Brayton + Rankine via HRSG — the most efficient conventional arrangement.

Practice

1. A Rankine plant has h_3 = 3,300 kJ/kg, h_4 = 2,350 kJ/kg, h_2 = 200 kJ/kg. Calculate its cycle efficiency (neglect pump work).
2. Explain, in energy terms, why regeneration raises efficiency even though bled steam does less turbine work.
3. A mill needs (a) emergency power within minutes and (b) large base-load power at the lowest fuel cost per kWh. Choose a plant type for each and justify in one sentence each.
4. Sketch the combined-cycle energy path and mark where the HRSG sits.

Answer key

1. Work = 3300 − 2350 = 950 kJ/kg. Heat in = 3300 − 200 = 3,100 kJ/kg. η = 950/3100 ≈ **30.6%**.
2. Bled steam's heat enters the feedwater instead of being rejected in the condenser, so the boiler adds less heat per kg; the fuel saving exceeds the small loss of turbine work.
3. (a) Diesel plant — starts in minutes and needs little water. (b) Combined-cycle (or large steam) plant — highest efficiency at large, steady output gives the lowest fuel cost per kWh.
4. Fuel → gas turbine (power 1) → hot exhaust → HRSG raises steam → steam turbine (power 2) → condenser. The HRSG sits between gas-turbine exhaust and the steam cycle, using exhaust heat as its only heat source.